ps4

Jenny Jia

2026-04-10

GitHub Repository: https://github.com/jennyjia319/gov51-ps4.git # 1 Setup and Exploration ## 1.1 Load packages and Data

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(estimatr)
library(modelsummary)
```

```
## Warning: package 'modelsummary' was built under R version 4.5.2
```

```r
library(dplyr)
card <- read.csv("data/card1995.csv")
```

1.2 Describe the Data

```r
head(card) # check to see if csv loaded
```

```
##   wage    lwage educ nearc4 nearc2 black smsa south exper expersq married age
## 1  548 6.306275    7      0      0     1    1     0    16     256       1  29
## 2  481 6.175867   12      0      0     0    1     0     9      81       1  27
## 3  721 6.580639   12      0      0     0    1     0    16     256       1  34
## 4  250 5.521461   11      1      1     0    1     0    10     100       1  27
## 5  729 6.591674   12      1      1     0    1     0    16     256       1  34
## 6  500 6.214608   12      1      1     0    1     0     8      64       1  26
##   fatheduc motheduc  IQ KWW
## 1       NA       NA  NA  15
```

```
## 2         8         8  93  35
## 3        14        12 103  42
## 4        11        12  88  25
## 5         8         7 108  34
## 6         9        12  85  38
```

```r
nrow(card)
```

```
## [1] 3010
```

```r
ncol(card)
```

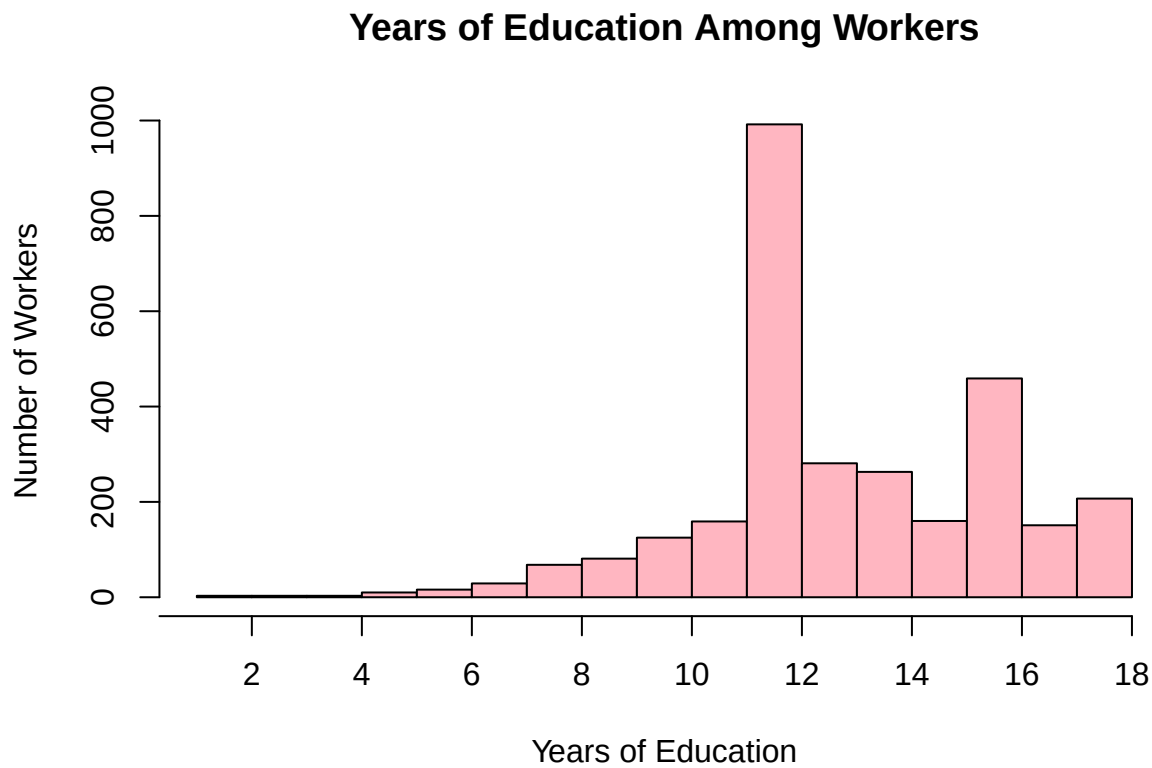
```
## [1] 16
```

There are 3010 observations and 16 columns. The unit of observation is a worker in the US.

1.3 Distribution of Years of Education

```r
hist(card$educ,
     main = "Years of Education Among Workers",
     xlab = "Years of Education",
     ylab = "Number of Workers",
     col = "lightpink",
     xaxt = "n")

axis(1, at = seq(0, 18, by = 2))
```




```r
mean(card$educ) # finding the mean
```

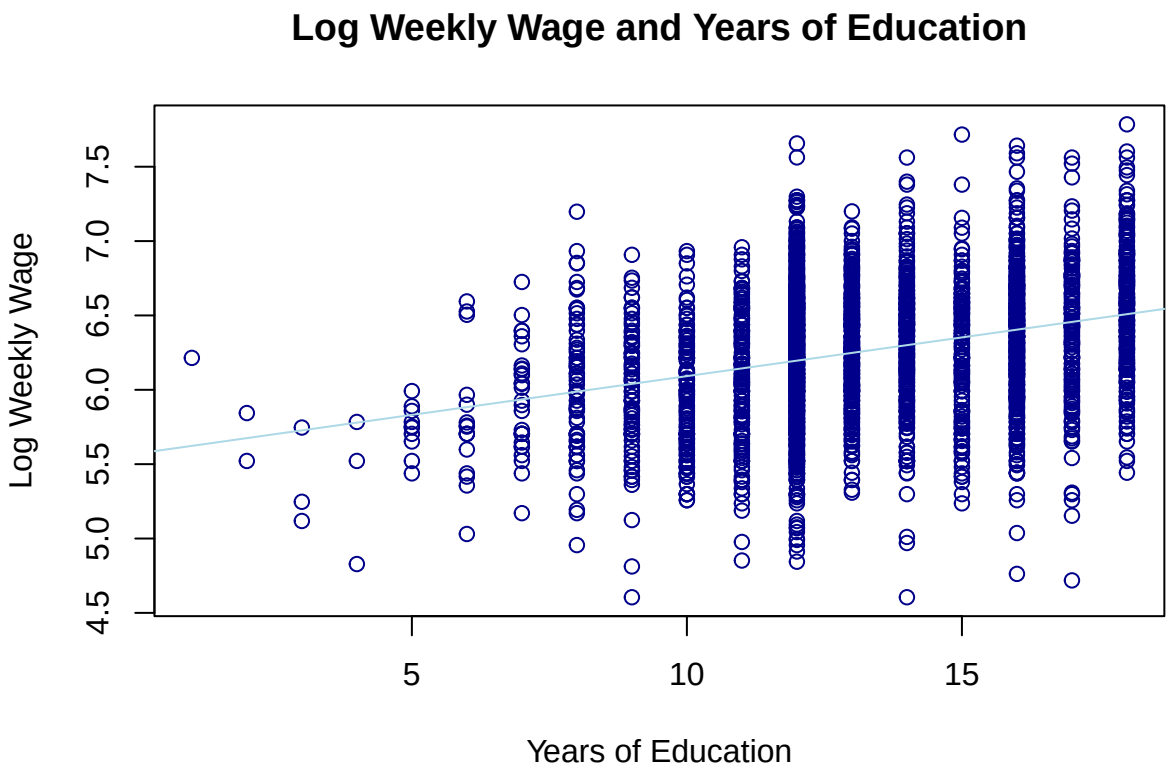
```
## [1] 13.26346
```

The distribution has a mean of 13.26346 years of education. There are two notable spikes, particularly around 12 years and 16 years. These are essentially the number of years it takes to complete high school and college, respectively, which tells us that there is a tendency to stop their education upon the completion of a degree.

1.4 The Naïve Relationship

```r
plot(card$educ, card$lwage,
     main = "Log Weekly Wage and Years of Education",
     xlab = "Years of Education",
     ylab = "Log Weekly Wage",
     col = "darkblue")

abline(lm(lwage ~ educ, data = card), col = "lightblue")
```



Yes, more education appears to be positively associated with higher log weekly wages. The regression line has a positive slope, which represents that each additional year of education is associated with an increase in log weekly wage.

1.5 Why OLS Might Be Misleading

The problem with the scatter plot above and the positive association interpretation is that this is a naive estimate. Since there may be omitted variable bias, where more motivated or wealthier workers tend to get more education and earn higher wages, it makes it hard to isolate the causal effect of education on its own.

2 The Instrument: College Proximity

2.1 What the Instrument Measures

```
mean(card$nearc4) # this can tell us the proportion, since nearc4 is binary
```

```
## [1] 0.6820598
```

68.20598% of respondents in this sample grew up near a 4-year college (where nearc4 = 1).

2.2 Relevance: Does Proximity Affect Education?

```r
first_stage <- lm_robust(educ ~ nearc4 + black + smsa + south + # pasted from pset
                           exper + expersq + married,
                         data = card, se_type = "HC2")
first_stage
```

```
##                 Estimate Std. Error     t value     Pr(>|t|)     CI Lower
## (Intercept) 16.946910655 0.16094303 105.2975739 0.000000e+00 16.631340582
## nearc4       0.327111413 0.08055446   4.0607488 5.016911e-05  0.169163750
## black       -0.945290863 0.08859855 -10.6693722 4.142485e-26 -1.119011027
## smsa         0.423328089 0.08521067   4.9680176 7.143751e-07  0.256250732
## south       -0.300175660 0.07857125  -3.8204260 1.359266e-04 -0.454234747
## exper       -0.432445687 0.03238761 -13.3521954 1.519509e-39 -0.495949899
## expersq      0.001409957 0.00170720   0.8258884 4.089332e-01 -0.001937447
## married     -0.074408252 0.01767034  -4.2109124 2.618347e-05 -0.109055482
##                 CI Upper   DF
## (Intercept) 17.262480728 2995
## nearc4       0.485059075 2995
## black       -0.771570699 2995
## smsa         0.590405447 2995
## south       -0.146116572 2995
## exper       -0.368941476 2995
## expersq      0.004757361 2995
## married     -0.039761021 2995
```

The coefficient on nearc4 is 0.327111413, the standard error is 0.08055446, the t-value is 4.0607488. Interpretation: Growing up near a 4-year college is associated with an average of 0.327111413 additional years of education, holding all other variables constant. I think this positive sign makes sense, since living near a college often means you're considered in-state, which lowers the cost tuition and commute, which thereby makes going to college more accessible.

2.3 Checking Instrument Strength

The t-value on nearc4 was 4.0607488, so F = the square of that, so 16.489681. Because this value > 10, it suggests that the instrument is strong. It's important that it's strong because it means that nearc4 explains a significant amount of variation in education, giving us enough reason to estimate the causal effect. A weak instrument would produce less precise 2SLS estimates with larger standard errors, and could introduce more bias.

2.4 Exclusion Restriction

(a) What must be true is that, when education is controlled for, nearc4 as an instrument must have no direct effect on lwage. This means the only reason people who grew up near a college earn higher wages is because they obtained more education, and not because of anything else about living near a college.

(b) An example of violation of the exclusion restriction is the fact that growing up near a college could be affected through socioeconomic status or university affiliation. Areas around colleges tend to attract more educated families, which means children could have grown up in a more elite and resource-abundant environment, which could boost wages without pursuing further education.

2.5 Balance Check

```r
card |>
  group_by(nearc4) |>
  summarise(across(c(black, smsa, south), ~mean(.x, na.rm = TRUE)))
```

```
## # A tibble: 2 x 4
##   nearc4 black  smsa south
##    <int> <dbl> <dbl> <dbl>
## 1      0 0.280 0.479 0.563
## 2      1 0.212 0.822 0.329
```

No, the groups are not balanced at all. Those who grew up near a college (nearc4 = 1) are less likely to be black (21.18850% vs 28.00418% for nearc4 = 0), are much more likely to live in a metropolitan area (82.222114% for vs 47.85789% for nearc4 = 0), and are less likely to live in the South (32.92742% vs 56.32184% for nearc4 = 0). This means that college proximity is not randomly-assigned, as it is correlated with all those variables. This means those variables could be confounders that cound affect both college proximity and wages, which means the instrument may violate the exclusion restriction.

3 The Wald Estimator

3.1 Compute the Reduced Form

```r
reduced <- lm_robust(lwage ~ nearc4 + black + smsa + south + exper + expersq + married,
                     data = card, se_type = "HC2")
reduced
```

```
##                 Estimate   Std. Error    t value     Pr(>|t|)     CI Lower
## (Intercept)  6.104783907 0.0369399692 165.262290 0.000000e+00  6.032353626
## nearc4       0.040854001 0.0160992779   2.537629 1.121090e-02  0.009287240
## black       -0.236173434 0.0183381490 -12.878804 5.575495e-37 -0.272130077
## smsa         0.196878652 0.0166562312  11.820120 1.526619e-31  0.164219841
## south       -0.146503379 0.0163740318  -8.947300 6.243086e-19 -0.178608866
## exper        0.040915614 0.0067799803   6.034769 1.787335e-09  0.027621724
## expersq     -0.001789075 0.0003227884  -5.542562 3.239634e-08 -0.002421984
## married     -0.038825630 0.0038391643 -10.113042 1.154798e-23 -0.046353296
##                 CI Upper   DF
## (Intercept)  6.177214187 2995
## nearc4       0.072420763 2995
## black       -0.200216791 2995
## smsa         0.229537464 2995
## south       -0.114397892 2995
## exper        0.054209503 2995
## expersq     -0.001156165 2995
## married     -0.031297964 2995
```

The coefficient on nearc4 is 0.040854001. In one sentence the interpretation is: Growing up versus not growing up near a four-year college is associated with a 4.0854001 percentage point increase in log weekly wages, holding all variables constant. This reduced form tells us how much wages change just from growing up near a college, without accounting for how much additional education the proximity caused.

3.2 Calculate the Ward Estimator by Hand

Wald estimator formula = reduced form coefficient on nearc4 / first stage coefficient on nearc4 = 0.040854001 / 0.327111413 = 0.12489323018

3.3 Interpret the Wald Estimator

One additional year of education caused by growing up near a college increases log weekly wages by 12.489323018 percentage points.

4 OLS vs 2SLS

4.1 OLS Estimate

```
ols <- lm_robust(lwage ~ educ + black + smsa + south + # pasted from pset
                   exper + expersq + married,
                 data = card, se_type = "HC2")
ols
```

```
##                 Estimate   Std. Error   t value     Pr(>|t|)     CI Lower
## (Intercept)  4.902202008 0.0712861999 68.767896 0.000000e+00  4.762427137
## educ         0.071498271 0.0036111667 19.799216 4.316524e-82  0.064417653
## black       -0.168717410 0.0173999796 -9.696414 6.533265e-22 -0.202834531
## smsa         0.172430654 0.0150653394 11.445521 1.017615e-29  0.142891194
## south       -0.127691620 0.0151740941 -8.415107 5.996475e-17 -0.157444322
## exper        0.071812773 0.0067378450 10.658122 4.654695e-26  0.058601501
## expersq     -0.001889027 0.0003149664 -5.997550 2.242922e-09 -0.002506599
## married     -0.033550125 0.0035795349 -9.372761 1.348435e-20 -0.040568721
##                 CI Upper   DF
## (Intercept)  5.041976879 2995
## educ         0.078578889 2995
## black       -0.134600289 2995
## smsa         0.201970114 2995
## south       -0.097938918 2995
## exper        0.085024046 2995
## expersq     -0.001271454 2995
## married     -0.026531529 2995
```

The coefficient on educ is 0.071498271. Interpretation: one additional year of education is associated with a 7.1498271 percentage point increase in log weekly wages, holding all variables constant.

4.2 2SLS Estimate

```
iv <- iv_robust(lwage ~ educ + black + smsa + south + # pasted from pset
                  exper + expersq + married |
                  nearc4 + black + smsa + south +
                  exper + expersq + married,
                data = card, se_type = "HC2")
iv
```

	OLS	2SLS
Years of Education	0.071	0.125
	(0.004)	(0.049)
Num.Obs.	3003	3003
R2	0.313	0.259

```
##                 Estimate   Std. Error   t value     Pr(>|t|)     CI Lower
## (Intercept)  3.988229477 0.8399956105  4.747917 2.151745e-06  2.341202728
## educ         0.124893231 0.0490790159  2.544738 1.098575e-02  0.028661238
## black       -0.118113004 0.0493318422 -2.394255 1.671531e-02 -0.214840728
## smsa         0.144007839 0.0305020895  4.721245 2.451737e-06  0.084200673
## south       -0.109013471 0.0230566857 -4.728063 2.371435e-06 -0.154222014
## exper        0.094925153 0.0223207958  4.252767 2.176073e-05  0.051159510
## expersq     -0.001965169 0.0003465968 -5.669900 1.564817e-08 -0.002644761
## married     -0.029532543 0.0051528211 -5.731335 1.095519e-08 -0.039635970
##                 CI Upper   DF
## (Intercept)  5.635256227 2995
## educ         0.221125225 2995
## black       -0.021385280 2995
## smsa         0.203815006 2995
## south       -0.063804928 2995
## exper        0.138690796 2995
## expersq     -0.001285577 2995
## married     -0.019429116 2995
```

The 2SLS coefficient on educ is 0.124893231. Interpretation: One additional year of education CAUSES a 12.4893231 percentage point increase in log weekly wages. Yes, this number essentially matches my Ward estimate that I got earlier of 0.12489323018

4.3 Side by Side Comparison

```r
modelsummary(list("OLS" = ols, "2SLS" = iv),
             coef_map = c("educ" = "Years of Education"), # pasted from pset
             gof_map = c("nobs", "r.squared"))
```

4.4 Why Do OLS and 2SLS Differ?

(a) 2SLS is larger at 0.125 versus 0.071 for OLS.
(b) First of all, years of education is self-reported, so respondents may not be accurately reporting them. This means the education variable contains noise, which means OLS is underestimating the true return on education. 2SLS avoids this because college proximity is correlated to how much education someone recieved, and not how accurately they reported it, which isolates the causal effect from confounders.
(c) The 2SLS estimate would be more credible as a causal estimate simply because it only uses variation in education that comes from proximity to a college, and not anything to do with their family background or other measurement errors.

4.5 Adding IQ As a Control

```r
ols_iq <- lm_robust(lwage ~ educ + IQ + black + smsa + south + exper + expersq + married, # (a) ols
                    data = card, se_type = "HC2")
ols_iq
```

```
##                 Estimate   Std. Error   t value      Pr(>|t|)     CI Lower
## (Intercept)  4.624820266 0.1106111619 41.811515 7.250324e-277  4.407898361
## educ         0.067568681 0.0050672360 13.334426  5.821735e-39  0.057631216
## IQ           0.002500262 0.0007491806  3.337329  8.609879e-04  0.001031028
## black       -0.120030879 0.0270219343 -4.441980  9.387896e-06 -0.173024170
## smsa         0.162782345 0.0184132040  8.840522  2.006318e-18  0.126671818
## south       -0.083180925 0.0183563649 -4.531449  6.195149e-06 -0.119179983
## exper        0.083230844 0.0092631008  8.985203  5.709939e-19  0.065064780
## expersq     -0.002350153 0.0004642726 -5.062012  4.518801e-07 -0.003260648
## married     -0.029240623 0.0044083448 -6.633016  4.194221e-11 -0.037885922
##                 CI Upper   DF
## (Intercept)  4.841742172 2051
## educ         0.077506145 2051
## IQ           0.003969496 2051
## black       -0.067037588 2051
## smsa         0.198892871 2051
## south       -0.047181867 2051
## exper        0.101396909 2051
## expersq     -0.001439659 2051
## married     -0.020595324 2051
```

```r
iv_iq <- iv_robust(lwage ~ educ + IQ + black + smsa + south + exper + expersq + married | # (b) 2sls
                   nearc4 + IQ + black + smsa + south + exper + expersq + married,
                   data = card, se_type = "HC2")
iv_iq
```

```
##                  Estimate   Std. Error    t value     Pr(>|t|)     CI Lower
## (Intercept)  4.1254833454 0.8039381042  5.1315932 3.143855e-07  2.548863208
## educ         0.1093899302 0.0669607951  1.6336414 1.024876e-01 -0.021928311
## IQ           0.0004228523 0.0034135214  0.1238757 9.014258e-01 -0.006271477
## black       -0.1233784512 0.0280423967 -4.3997114 1.139525e-05 -0.178372993
## smsa         0.1536399386 0.0239013796  6.4280783 1.602608e-10  0.106766434
## south       -0.0857686314 0.0190205986 -4.5092498 6.872948e-06 -0.123070332
## exper        0.1035065280 0.0339813212  3.0459830 2.348745e-03  0.036865035
## expersq     -0.0027889965 0.0008630306 -3.2316312 1.250362e-03 -0.004481504
## married     -0.0274086703 0.0053001348 -5.1713157 2.550570e-07 -0.037802878
##                 CI Upper   DF
## (Intercept)  5.702103483 2051
## educ         0.240708171 2051
## IQ           0.007117182 2051
## black       -0.068383910 2051
## smsa         0.200513443 2051
## south       -0.048466930 2051
## exper        0.170148021 2051
## expersq     -0.001096489 2051
## married     -0.017014463 2051
```

(a) The OLS coefficient on educ is 0.067568681 with IQ added, which is lower than 0.071498271 (no IQ). This tells us that there is slight mobility bias in the original estimate without IQ, so the estimated return on education was capturing differences in ability and not just the causal effect of education alone.
(b) The 2SLS coefficient on educ is 0.1093899302 with IQ added, which is lower than 0.124893231 (no IQ). It falls, but only modestly. This tells us that ability does not play a major role in biasing the iv estimate.
(c) When we add IQ to both OLS and 2SLS and both coefficients only decrease mildly, we take away that ability bias is not the main source of the gap between OLS and 2SLS. So, the difference might then be explained by measurement error (because education is self-reported).

5 Interpretation and LATE

5.1 Who are the Compliers?

The compliers are the respondents that pursued further education because they grew up near one, and they would not have pursued it if they didn't grow up near one (where nearc4 = 0). This would've been people that would have been impacted by the cost and logistics considerations of attending college, and were in-between attending college or not, college proximity being what made attending college possible. So, someone wealthy or motivated children would have attended college anyways, regardless of proximity, so those people would not count as a complier group.

5.2 Write the LATE interpretation

Among workers whose years of schooling were influenced by whether they grew up proximate to a 4-year college, obtaining one additional year of education causally increases log weekly wages by 12.4893231 percentage points.

5.3 External Validity

Building off my answer to 5.1, the compliers are people who were on the "cusp" of attending college, who only pursued more education because they happened to live close, and not that they would've gone regardless. We infer that there would be higher returns for this complier group, as the proximity provided them the boost necessary to pursue education. This means LATE is likely higher than ATE, because ATE would include the people that would have attended college regardless of where they lived and would have lower returns to additional education.